

Fuel Monitoring System based on IoT: Overview and Device Authentication

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Abstract— In recent years, more and more devices, such as smart mobile devices, smart tags, sensors, or chips have been connected to the Internet. All these devices are part of the Internet of Things, a technology that has experienced rapid development, along with technologies such as cloud computing or big data. In this paper we propose a system for monitoring fuel consumption for a fleet of locomotives, using Raspberry Pi devices to which sensors are connected, in the context of the Internet of Things. We also emphasize the authentication of Raspberry Pi devices inside the system and present how data consumption data are transmitted between the components of the system.

Keywords—Internet of Things, authentication, Raspberry Pi, fuel monitoring

I. INTRODUCTION

Technology has advanced rapidly in recent years. In the last two years alone, given the context around the world, many companies and organizations have shifted their business to the online environment, and online communication has gained an important role in maintaining human interaction. The adoption of cloud platforms has seen a considerable increase in all areas. Also, the connection of different types of devices to the internet has experienced a significant increase. All of these Internet-connected devices are embedded in the Internet of Things (IoT). IoT encompasses any device that is connected to the Internet, from smart mobile devices to sensors, chips, or microcontrollers. In [1] is shown that only in 2018 there were 50.1 billion connected IoT devices included in smart vehicles, smart homes, or even further smart cities. In the field of transport, IoT has a high potential, because by connecting different types of sensors to an Internet network, important data can be transmitted regarding the condition of the vehicle itself, whether it is cars, trains, ships, or even planes. In addition to information about the vehicle, information about the environment in which the vehicle travels can also be collected through sensors, thus helping to apply optimizations to the level of fuel consumed, or choosing routes based on weather conditions, for example.

In this paper, we present a system for monitoring fuel consumption for a fleet of locomotives in the context of the Internet of Things. Also, we present the authentication process of the IoT devices within the system. We oriented to locomotives because these can use diesel or electricity as fuel, therefore there are two scenarios covered. Having such a system, then it can be easily ported to vehicles that use only electricity, or diesel respectively as fuel. The contributions of the paper can be summarized, as follows: (1) designing a system that monitors the fuel consumption for a fleet of locomotives; (2) secure authentication of the IoT devices within the proposed system; (3) potential for practical

implementation as all components of the proposed system are easily achievable. Moreover, the work can be extended for further scenarios. For example, gathering knowledge about consumption tendency, different statistical models can be applied such that the consumption to be optimized and to become more environmentally friendly.

The rest of the paper is organized as follows: in section 2 we present background information related to the IoT domain and the authentication of an IoT device within a network/system and present related work with the proposed system. Section 3 is dedicated entirely to the proposed system, as follows: we firstly present the components of the system and their roles, then we present the authentication mechanism for the IoT devices, and lastly, we show the strengths and the advantages of the proposed system. In the last section, we present the conclusions of the paper and further research directions.

II. BACKGROUND PRELIMINARIES

A. Background

In this subsection, we present the main aspects of the mechanisms used in our system.

Authentication. IoT devices help a lot to improve processes because the data collected from them can be analyzed to find patterns, make recommendations or predict incidents before they take place. However, one very important thing is the security associated with such devices, and for this, the IoT architecture is the key characteristic. In [2], [3] the authors propose an organization of IoT architecture on 3 layers: application, network, perception. Other studies [4] propose five layers of the IoT architecture: business, application, processing, transport, perception. Authentication of the IoT devices can be made mainly at the perception level or application level [5]. In [5] the authors present a comprehensive study about authentication mechanisms for IoT devices, from where the main classes of authentication are: hardware-based authentication, token-based authentication, and procedure-based authentication.

OTP (One-Time Password). OTP represents a string of characters generated randomly, which is used just one time for authentication in a single login session. Usually, the OTP is generated by a specialized server. There are two types of OTP [6], [7]: time-synchronized OTP and OTP based on mathematical algorithms. The first category is valid only for a short period of time (for example, a few minutes), then it expires. If the OTP was not used in a specific interval of time, then it becomes invalid. The second type of OTP can be used anytime after it was generated. However, in practice, this

category has a limited time of use, too, but is much longer (for example, 12 or 24 hours).

Challenge-response mechanism. This is a protocol between two parties, the challenger, and the responder, in which the challenger challenges the responder usually with a value, and the responder must reply with a correct corresponding value. If the responder's answer is correct, then it can proceed further. Usually, the challenge-response mechanism is used in authentication processes. The challenge-response mechanism is a preferred authentication mechanism for IoT devices [8], [9], [10]. Challenge-response mechanism can be used with OTPs (for example, [11]).

Hash functions. These are one of the most used cryptographic primitives met in almost any security system or application. A hash function takes the input of an arbitrary length (it can be anything, from string values to different types of files) and outputs a string with a specific length, called a hash value. Ideally, the hash value should be unique for every input, but since the input space is infinite and the output space is finite there will be situations in which two different input values will have the same output, value called *collision*. The properties of hash functions are pre-image resistance, second pre-image resistance, collision resistance. These properties ensure the uniqueness of the output. Even with a slight change of the input (for example, one bit), the output should also change. Hash functions are very good in integrity checking. Examples of hash functions are MD family, SHA family, RIPEMD, Keccak [12].

B. Related Work

IoT devices are widely used in the vehicle industry, in various use cases. For example, they can be used for real-time monitoring of vehicle locations to manage traffic, for retrieving information from different sensors in optimization scenarios, and so on. In this subsection, we will emphasize related work included in the second category. In [13] the authors propose a fuel monitoring system for vehicles (car or bike) for individual use, based on Raspberry Pi devices. Also, the proposed system includes location tracking and a mobile application in which the user can check information about fuel consumption. The system proposed in [14] is also designed for individual use. Besides fuel monitoring, the solution from [14] provides information regarding the nearest gas stations in the vehicle's proximity and the prices for each gas station. In [15] the authors propose a similar system, but it can be adapted to a fleet of automobiles. In [16] the authors use IoT and blockchain for the system that monitors fuel consumption. Also, the authors prove the efficiency of their proposed system and show it is secure against collusion attacks. On the other hand, in [17] the authors presents the challenges raised by monitoring and control systems with a focus on elective vehicles.

III. THE PROPOSED SYSTEM

The proposed system is designed for a fleet of locomotives, but it can be adjusted for different types of fleets. The motivation for using locomotives for the proposed system lies in the fact that these can be diesel locomotives or electrical locomotives, covering two different types of vehicles in the same system. In the remaining of the paper, we refer to vehicles as locomotives. In this section, we focus on the overview of the system, presenting its components and workflow, and on the authentication process for the IoT devices.

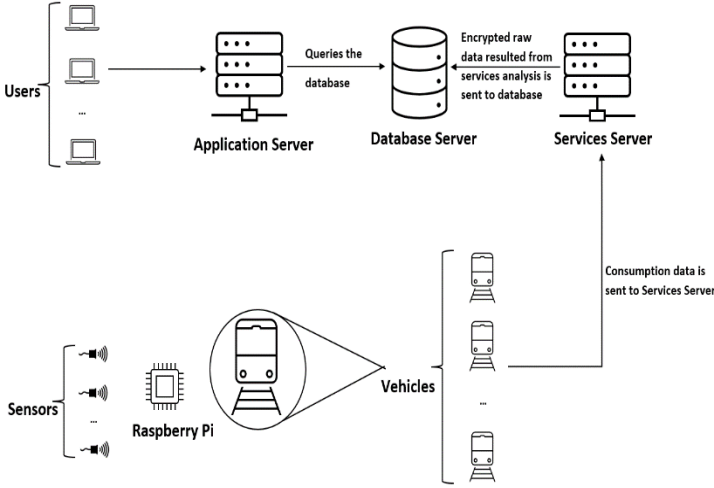
A. Overview of the Proposed System

To have a clear overview of the system, we first present its components. Also, Fig. 1 shows graphically the components of the system and the relation between them.

- *Vehicles.* Each vehicle is equipped with one Raspberry Pi device, to which several sensors are attached. At least one sensor should be attached, namely the sensor that reads the level of diesel, or the sensor that reads the electricity, respectively. However, other sensors can be attached, such as a temperature or humidity sensor. A list of the most used sensors connected to Raspberry Pi can be found in [18]. Also, the Raspberry Pi device should contain a cellular adapter that enables it to use the SIM card. The device is programmed to collect consumption raw data into a buffer and send the raw data within the buffer to the services' server at a specific amount of time (for example, at 24 hours). The time interval at which the raw consumption data is transmitted to the server must be configurable via an SMS code.
- *Server for services.* This server receives the raw data from vehicles. Here, processing services should be installed and configured, such that the raw data (from electrical and diesel vehicles) be processed and then transmitted to the database server. After the data is processed, it is sent to the database server, where it will be stored.
- *Server for databases.* This server hosts the database in which the information about consumption is collected. It received the data from the service server.
- *Server for application.* This server hosts the application through which the owner of the fleet can see the information about consumption. It can include tabular or graphical reports that show the evolution of fuel consumption for both diesel and electric vehicles. If the Raspberry Pi installed on the vehicles contains other sensors, then the application can include many other types of reports (humidity, temperature, etc). The application server interrogates the database in order to generate the reports.
- *Users.* These are individual clients of the application. A user can be an employee of the vehicle fleet company or the owner(s) of the vehicle fleet company. Depending on the type of user account, a user can access different reports. However, an admin user should exist to manage the access, user account, types of available reports, etc.

Further, we present how the system works. The central part of the system is the Raspberry Pi devices installed on each vehicle. Every time a device is installed on a vehicle, it is retained in the database a pair of the form (ID_V, ID_R) , where ID_V is the unique identifier of the vehicle and ID_R is the unique identifier of the Raspberry Pi device. Also, the Raspberry Pi is programmed to know its unique identifier. To each such device, different sensors are attached, but the most important is the sensor for fuel consumption measuring (diesel or electricity) and the cellular adapter for the SIM card. The Raspberry Pi is programmed to collect real-time consumption data into a memory buffer, then, at specific intervals of time, the data is transmitted to the services' server. Here two aspects are important: configuration and authentication.

Fig. 1. Overview of the proposed system and the workflow



The interval of time can be configured by an administrator of the system through an SMS code, therefore on the Raspberry Pi, a configuration service should run. When the admin sends the SMS code, the configuration service checks and validates the phone number that has sent the code. This can be done by checking within a list of authorized phone numbers. If the sender number is valid, then the service configures the interval of time at which the consumption data is transmitted to the services' server. The second important thing is the authentication of the device. Each time the Raspberry Pi sends the consumption data to the server, it should firstly authenticate within the system. If it proves correctly its identity, then it can proceed with the data sending process (the process of authentication will be detailed in the next subsection). Before sending the consumption data, the service that runs on the device will encrypt them, and then it sends the encrypted data to the services' server via the 3g/4g/5g network connection enabled through the SIM card. Once the encrypted data arrives on the server for services, these are decrypted and then processed by the processing service. Further, processed data is encrypted and transmitted to the database server. Once the encrypted data arrives on the database server, these are decrypted by a specific service and stored in the database. Further, the application server may interrogate the database in order to provide to the user meaningful data. Note that each time data is transmitted between components, firstly the data is encrypted, and secondly, the transmission should be made over a secure channel. In this paper, we will not focus on the encryption systems used to encrypt the data or the secure communication channel. However, data can be encrypted using the RSA (Rivest Shamir Adleman) encryption system [19], for example, or another lightweight encryption algorithm that can be handled by the processing power of the IoT device.

B. The Authentication of the IoT Device

In the context of a network of IoT devices, one of the most important things is that the device that communicates with the rest of the devices or other components of the network to be valid and legitimate. By valid, we refer to an uncorrupted device and authorized device to link to the network and communicate with others. A level of such security can be accomplished through the authentication of the IoT device within the network. In our system, the IoT device is the Raspberry Pi device. The authentication process for the Raspberry Pi device works as follows:

1. The device contacts the services' server.
2. The services' server generates an OTP, encrypts it with a specific encryption algorithm, and sends the value to the Raspberry Pi device. OTPs are useful because these can be used just one time. Once it was used, an OTP becomes invalid and cannot be used in subsequent sessions or authentication processes.
3. The Raspberry Pi device decrypts the encrypted OTP and computes the following value:

$$v = h(\text{OTP} \otimes \text{ID}_R) \quad (1)$$

where h is a hash function (one of the best hash functions is SHA-512), OTP is the value received from the server and ID_R is the unique identifier of the Raspberry Pi device on which Raspberry Pi knows. The value v is then encrypted and transmitted to the services' server.

4. The server makes the same computations:

$$v' = h(\text{OTP} \otimes \text{ID}_R) \quad (2)$$

where h is the same hash function used by the Raspberry Pi, OTP is the OTP value generated by the server and ID_R is the unique identifier of the Raspberry Pi device on which the server may retrieve from the database.

5. The server decrypts the value received from the Raspberry Pi (v).
6. If $v=v'$, then the server allows the Raspberry Pi device to connect to the network and send consumption data, otherwise the device is not allowed to connect to the network.

Security. Each time the data is transmitted between components of the system, they are encrypted before transmission. We are not focusing on this paper on how data is encrypted because there is a wide range of encryption techniques that can be used (RSA, elliptic curve-based cryptosystems, etc). Moreover, depending on the components involved it can be used even different cryptosystems within the system. However, an important part is the authentication of the Raspberry Pi device within the system, before transmitting data. Here, the following procedures contribute to the security of the authentication:

- *OTP.* Using an OTP each time when the device is programmed to transmit consumption data, the system is protected against reply attacks, meaning that if an adversary succeeds to seize an OTP, then it cannot use it in further authentication processes because the OTP can be used just one time, and becomes invalid once it was used. Moreover, OTPs are generated based on randomness, therefore, it is very difficult to predict an OTP value at an arbitrary connection request. Also, the OTP can be programmed to be valid for a specific short period of time. If the device does not respond in that interval of time, then the connection is dropped.
- *Hash.* Even an adversary manages to capture and decrypt the encrypted hash value sent by the Raspberry Pi to the server, and if tries to alter the data, the server will detect the alteration after it makes its own computations to obtain the value v' due to the hash

function. Using a hash function, the integrity of the message is ensured.

- *Challenge-response authentication.* The server challenges the Raspberry Pi device with the OTP because the server expects a value from the Raspberry Pi based on that OTP. On its side, Raspberry Pi responds with the computed value v .
- *Individual communication.* This is related to the communication within the system. By the system design, the Raspberry Pi devices do not communicate between them. Instead, each Raspberry Pi communicates only with the service's server. In the case of Raspberry Pi corruption, it makes it easy to isolate it and eliminate it from the system, a fact that does not affect the rest of the devices.

The systems presented in related work are oriented towards cars or fleets of cars, rather than trains or similar vehicles and mainly the authentication is designed for the users of the system rather than the authentication of the IoT devices. In addition, most are intended for individual users. In Table 1 we compare the techniques used in different authentication mechanisms for the IoT devices with our and show the target of the proposed systems.

TABLE I. AUTHENTICATION MECHANISMS COMPARISON

Work	Authentication mechanisms	Designed for
[20]	Unclonable source authentication signature	Fleet Management Systems (transport, distribution, and logistics)
[21]	Token generated by the HTTPS protocol headers	Intelligent Transportation Systems (route compliance, schedules, speed control)
[22]	Blockchain-based authentication	Autonomous Vehicles System
[23]	Fog Computing and Software-Defined Networking	Intelligent Cars System
Ours	Challenge-response authentication with OTP	Fuel Consumption Monitoring for a Fleet of Locomotives

IV. CONCLUSIONS

In this paper, we proposed a system for monitoring the fuel consumption for a fleet of locomotives, which support diesel consumption and energy consumption. One of the most important components within the system is the IoT device, in the form of a Raspberry Pi to which sensors are connected. We also presented the authentication mechanism of the IoT device and highlighted the security of the authentication process. As further research, we will extend the capabilities of the proposed system, by including geo-location tracking in real-time. Also, we will research the machine learning algorithm that can provide insightful information from data analysis such that to improve the fuel consumption in relation to different parameters, such as the engine's state or weather conditions.

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